MODULE 4

MBRAMOS

Networking and Communications

CIDR Notation & Subnetting

Classless Inter-Domain Routing (CIDR) Notation

Go back to Slide #2 Binary-to-Decimal Conversion

128	64	32	16	8	4	2	1	Total
1	1	1	1	1	1	1	1	**255**

192.168.30.4	/24	255.255.255.0	/24

8	8	8	0
255	255	255	0

/24

In the early 1990s, the internet was a victim of its own success. The original method of allocating IP addresses known as Classful Addressing was dangerously inefficient and threatened to break the global routing system. **Classless Inter-Domain Routing (CIDR) was introduced in 1993 to prevent the internet from running out of space and to stop core routers from being overwhelmed.**

What is Subnet Mask?

Go back to Slide #5 Classes of IP Addresses

Class	1st Octet	Number of Hosts	Subnet Mask
Class A	1 - 126	16.7 Million	255.0.0.0
Class B	128 - 191	65 Thousand	255.255.0.0
Class C	192 - 223	254	255.255.255.0

192.168.30.4	/24	255.255.255.0	/24

8	8	8	0
255	255	255	0

/24

Subnet mask represent the exact same information as CIDR Notation, they just show it in different "languages." as a simple explanation. Think of a Subnet Mask as the formal, long-form name (e.g., Jonathan) and CIDR Notation as the nickname (e.g., Jon). They both point to the same person, but one is much faster to write.

CALCULATING THE SUBNET MASK & CIDR (CLASS C)

To speed up subnet mask calculations, I created a chart you should memorize. The highlighted rectangles will be used on the next slide.

128	64	32	16	8	4	2	1	TOTAL	REFERENCE
1	0	0	0	0	0	0	0	128	[1]
1	1	0	0	0	0	0	0	192	[2]
1	1	1	0	0	0	0	0	224	[3]
1	1	1	1	0	0	0	0	240	[4]
1	1	1	1	1	0	0	0	248	[5]
1	1	1	1	1	1	0	0	252	[6]
1	1	1	1	1	1	1	0	254	[7]
1	1	1	1	1	1	1	1	255	[8]

Subnet	1	2	4	8	16	32	64	128	256
Hosts	256	128	64	32	16	8	4	2	1
CIDR	/24	/25	/26	/27	/28	/29	/30	/31	/32

8 +	8 +	8 +	2	=	/26
255	255	255	192		

SUBNETTING A CLASS C NETWORK

This means only /24 and above

Question:

Your supervisor gave you an IP addresses of **192.168.4.0** and asked you to chop it into **three smaller, isolated pieces**. Your job is to use subnetting to create separation between the **Front Desk, HR, and Accounting** so their data stays private and organized.

Subnet	1	2	4	8	16	32	64	128	256
Hosts	256	128	64	32	16	8	4	2	1
CIDR	/24	/25	/26	/27	/28	/29	/30	/31	/32

Network ID (NID)	First IP (FIP)	Last IP (LIP)	Broadcast IP (BIP)
0	1	62	-1 63
-1 64 +1	65	126	-1 127
-1 128 +1	129	190	-1 191
-1 192 +1	193	254	-1 255

TOTAL	REFERENCE
128	[1]
192	[2]
224	[3]
240	[4]
248	[5]
252	[6]
254	[7]
255	[8]

SUBNETTING BENEFITS

1. **Minimized Broadcasting:** Since a broadcast sends data to every computer in a network, reducing its reach results in better overall performance. Think of it like having fewer cars on the road, less cars on the highway, better traffic flow
2. **Simplified Management:** Issues can be easily isolated within a specific subnet. For example, if a device in the HR department is phished and a self-spreading virus is installed, the infection will be contained within that subnet range rather than spreading to the entire company.
3. **Enhanced Security:** Host traffic can be strictly filtered using IP Access Lists. Employees in Accounting can only access accounting resources, while Marketing stays within their own lane, preventing unauthorized access to sensitive data. Only the Network Administrator (usually the IT department) retains access to all segments.